

Case Study

Food Delivery Operations & SLA Performance Analysis

Problem Statement

The operations team at a food delivery company has observed a steady increase in customer complaints regarding delayed deliveries but lacks clear visibility into the primary causes of these delays. Although thousands of food orders are delivered every day, management cannot confidently determine whether delivery delays are mainly driven by city-specific operational issues, weather conditions, traffic congestion, vehicle type, or delivery rider performance.

Without these insights, operational decisions rely heavily on generic process improvements rather than targeted, data-driven interventions. This results in continued SLA breaches, declining customer satisfaction, higher compensation costs, and inefficient rider allocation.

The objective of this analysis is to identify the key operational factors affecting delivery performance, evaluate where service-level agreements (SLAs) are breaking down, and provide actionable recommendations to improve delivery efficiency and customer experience.

Data Overview

- **Source:** Food Delivery Dataset
- **Records:** ~45,500 Delivery Orders
- **Tools Used:** Excel, MySQL, Power BI

The dataset contains delivery rider information, order timestamps, delivery duration, traffic conditions, weather conditions, vehicle type, city, rider ratings, and operational attributes used to evaluate delivery performance.

Tool Stack

Excel (Data Cleaning) → **MySQL** (Business Analysis) → **Power BI** (Interactive Dashboard)

Business Questions

1. Which city records the highest average delivery time and late delivery rate?
2. Which vehicle type delivers orders the fastest?
3. How much does traffic density increase average delivery time?
4. Which time of day experiences the highest percentage of late deliveries?
5. Which weather conditions have the greatest impact on delivery performance?
6. Who are the top-performing and lowest-performing delivery riders based on delivery time?

Data Cleaning (Excel)

Performed data validation and quality checks before analysis.

Cleaning Steps

- Cleaned **Delivery Time** by extracting numeric values from the text field.
- Replaced missing values in **Delivery Person Ratings** using the column average.
- Removed unrealistic delivery rider ages (below 18 years and above 65 years).
- Standardized **Weather Conditions** by removing unnecessary text prefixes.
- Verified missing values and standardized categorical values.
- Created **Time of Day** column based on order time.
- Created **Late Delivery Status** column using a 40-minute SLA threshold.
- Created **Delivery Speed Band** (Fast, Normal, Slow, Very Slow) for delivery performance analysis.
- Verified data types and removed invalid records.

Analysis (SQL)

Key Findings

1. Which city has the worst delivery performance?

Urban has the worst avg delivery time at **23 min** and a late rate of **4.08%** — significantly above the company average of **26.29 min**.

City	total_deliveries	avg_delivery_time_min	fastest_delivery	slowest_delivery	late_deliveries	late_delivery_rate_pct
Semi-Urban	152	49.71	44	54	152	100.00
Metropolitan	32671	27.31	10	54	3192	9.77
Urban	9740	23.00	10	54	468	4.80
NaN	1138	22.07	10	54	35	3.08

2. Which vehicle type delivers the fastest?

Electric Scooter is the fastest at avg **24.41 min** while **Motorcycle** is slowest at **27.60 min** — a **3** minute gap that compounds across thousands of deliveries.

Type_of_vehicle	total_deliveries	avg_delivery_time_min	avg_rider_rating	late_deliveries	late_rate_pct
electric_scooter	3545	24.41	4.66	159	4.49
scooter	14626	24.45	4.65	703	4.81
motorcycle	25523	27.60	4.63	2985	11.70
bicycle	7	28.71	6.00	0	0.00

3. How much does traffic affect delivery time?

Heavy traffic adds **30 minutes** on average compared to **low traffic** — a **46%** increase in delivery time, making traffic the single biggest operational variable

Road_traffic_density	total_deliveries	avg_delivery_time_min	min_delivery_time	max_delivery_time	late_rate_pct
Jam	13725	31.14	10	54	19.86
High	4296	27.20	10	54	6.17
Medium	10628	26.71	10	49	6.04
NaN	53	23.94	11	48	3.77
Low	14999	21.29	10	44	1.41

4. Which time of day experiences the highest delays?

Evening (5–8pm) is the peak delay window — **13.22%** late rate vs **1.36%** in the morning — the worst time to have the fewest riders on the road.

time_of_day	total_deliveries	avg_delivery_time_min	late_rate_pct	avg_rider_rating
1. Morning (6-11am)	7674	21.28	1.36	4.67
2. Afternoon (12-4pm)	4029	25.63	4.00	4.67
3. Evening (5-8pm)	17797	29.21	13.22	4.62
4. Night (9pm-5am)	14148	25.53	8.68	4.62

5. Which weather condition causes the most SLA breaches?

Fog causes the highest SLA breach rate at **15%** — but even in sunny conditions **4.55%** of deliveries are late, confirming weather is not the only driver.

weather	total_deliveries	avg_delivery_time_min	sla_breaches	sla_breach_rate_pct
Fog	7440	28.93	1122	15.08
Cloudy	7288	28.92	1079	14.81
Windy	7223	26.11	457	6.33
Sandstorms	7245	25.88	450	6.21
Stormy	7374	25.87	415	5.63
Sunny	7078	21.85	322	4.55
	53	23.94	2	3.77

6. Which delivery riders perform best?

The gap between the top and bottom rider is **8 minutes** avg delivery time — the bottom 10 riders account for a disproportionate share of all late deliveries

Top 10

Delivery_person_ID	total_deliveries	avg_delivery_time_min	avg_rating	late_rate_pct
CHENRES03DEL01	54	30.96	4.55	14.81
CHENRES14DEL01	51	30.27	4.59	17.65
VADRES11DEL02	64	29.64	4.60	18.75
VADRES01DEL01	57	29.54	4.59	19.30
COIMBRES16DEL02	55	29.53	4.59	16.36
JAPRES09DEL03	58	29.31	4.60	15.52
JAPRES010DEL01	54	29.31	4.61	12.96
PUNERES14DEL01	54	29.28	4.60	12.96
RANCHIRES17DEL03	40	29.20	4.64	15.00
COIMBRES05DEL02	52	29.02	4.62	15.38

Bottom 10

Delivery_person_ID	total_deliveries	avg_delivery_time_min	avg_rating	late_rate_pct
COIMBRES07DEL03	40	22.23	4.76	2.50
PUNERES010DEL03	41	22.56	4.63	4.88
MUMRES12DEL02	54	22.70	4.69	1.85
PUNERES07DEL02	59	22.71	4.69	1.69
VADRES03DEL03	43	22.79	4.65	2.33
BANGRES17DEL03	42	23.07	4.63	4.76
RANCHIRES09DEL03	44	23.09	4.63	2.27
BANGRES15DEL02	57	23.18	4.58	3.51
HYDRES18DEL02	56	23.20	4.71	1.79
BANGRES15DEL03	44	23.23	4.69	4.55

Dashboard (Power BI)



Recommendations

1. Optimize Rider Allocation During Peak Hours

Increase rider availability during evening peak hours in cities experiencing the highest late delivery rates to reduce SLA breaches.

2. Improve Vehicle Assignment Strategy

Assign the fastest-performing vehicle types to high-priority and express deliveries while utilizing slower vehicles for non-urgent orders.

3. Implement Dynamic ETA Predictions

Incorporate real-time traffic and weather conditions into estimated delivery time calculations to improve delivery accuracy and customer expectations.

4. Establish Rider Performance Monitoring

Develop monthly rider performance scorecards using delivery time, customer ratings, and SLA compliance to identify training opportunities and reward high-performing riders.

5. Introduce Weather-Based Operational Planning

Implement weather-triggered operational strategies, including proactive customer communication and temporary SLA adjustments during adverse weather conditions.

6. Improve City-Level Resource Planning

Use historical delivery patterns to dynamically allocate riders based on city demand, traffic density, and peak delivery periods, reducing delivery delays and improving operational efficiency.